Alert-Budgeted Clinical Early Warning: Conformal Risk Control for Stateful Deterioration Alarm Policies

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Abstract

Clinical early warning systems are evaluated almost exclusively by discrimination, yet their failure in deployment is operational rather than statistical: intensive care units generate alarm volumes far beyond what clinical staff can absorb, and the resulting alarm fatigue erodes the very safety the systems are meant to provide. This paper treats the alarm budget, defined as the tolerated number of alarms per patient-day, as a first-class design constraint rather than a post hoc threshold adjustment. We formalise early warning as a stateful alarm policy whose refractory window and escalation tier are explicit control variables, and we show that the resulting temporal dependence invalidates timestep-level conformal calibration. We therefore develop a patient-level blocked conformal risk control procedure that restores finite-sample validity for dependent alarm sequences and yields an upper bound on the probability that the deployed population alarm rate exceeds the operational budget. The method is instantiated on irregularly sampled intensive care trajectories using two open, multi-hospital cohorts, and is evaluated under cross-hospital transfer, prevalence shift, and observation sparsification. The AI contribution is a calibration procedure that provides distribution-free control of alarm rate for policies with internal state; the engineering contribution is an operating-point selection mechanism that lets a monitoring system be specified by its alarm budget and then certified against it. The evaluation protocol compares all methods at matched alarm budgets rather than at matched thresholds, reports detection lead time and net benefit alongside discrimination, and quantifies the empirical budget violation rate over

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repeated calibration splits.

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1. Introduction

Deterioration events in hospitalised patients unfold over hours, and the physiological trajectory that precedes them is recorded long before the event itself is recognised. This temporal structure has motivated a substantial literature on data-driven early warning, ranging from static severity scores to sequential models that forecast deterioration trajectories from electronic health records [1] and compact, efficiency-aware networks that predict sepsis onset from structured time series [2]. Reported discrimination is high and improving, with in-hospital mortality and onset prediction now routinely exceeding the performance of classical scoring instruments [3]. The implicit promise of this line of work is that a sufficiently accurate model, once installed at the bedside, will shorten time to intervention.

That promise has not been realised at scale, and the reason is operational rather than statistical. A prediction model produces a probability at every observation time for every patient; converting those probabilities into alerts requires choosing a threshold, and the threshold that maximises discrimination on a retrospective cohort typically generates an alert volume that a clinical unit cannot absorb. Recent methodological reviews of adverse-event prediction in intensive care document that reported performance is dominated by discrimination metrics, while alert burden, time-to-alert, and calibration under deployment conditions are reported inconsistently or not at all [4]. The same imbalance appears in reviews of deep learning for multivariate clinical time series, where architectural novelty is emphasised far more than the operating characteristics of the resulting monitoring system [5, 6]. Consequently, two models with similar area under the receiver operating characteristic curve can differ by several-fold in the number of alarms they emit, and the literature offers no principled way to compare them at a common operational cost.

The dominant technical response has been to improve the backbone that produces the risk score. Because clinical observations are irregularly sampled and heterogeneously missing, a rich family of methods models obser-

vation times explicitly, including continuous-time and reliability-aware temporal aggregation [7, 8], graph formulations over irregular observations [9], and masked-attention models that exploit informative missingness for deterioration prediction in general wards [10]. Transfer-oriented variants further align feature spaces across cohorts so that a model trained in one setting can be reused in another [11], and dynamic temporal graphs incorporate clinical calculators directly into the prediction pathway [12]. These contributions are complementary to ours and are adopted here as backbones, but they leave the operating point untouched: the mapping from scores to alerts is still a threshold selected after training, and the cost of that choice is still unmeasured.

A separate literature has begun to attach reliability guarantees to clinical predictions. Selective prediction and calibration have been analysed systematically for multimodal clinical classification [13], reliability estimation has been extended beyond output-space calibration for time-series classifiers [14], and conformal methods have been used to equip physiological regression with statistically calibrated uncertainty [15]. These treatments share an assumption that is violated by early warning systems in operation: they calibrate a decision rule applied independently to exchangeable examples. An alerting system is not a sequence of independent decisions. Once an alarm fires, clinical practice and good engineering alike suppress subsequent alarms for a refractory period, so the decision at time t depends on the decision at $t-1$. This internal state breaks the exchangeability that standard conformal calibration relies on, and the resulting guarantee is silently invalid precisely in the regime where alarms cluster.

This paper closes that gap by making the alarm budget an explicit constraint and calibrating the policy that enforces it. Our contributions are as follows.

- We formulate early warning as a stateful alarm policy in which the refractory window and escalation tier are control variables, so that alarm volume becomes a design parameter of the monitoring system rather than an emergent property of a threshold.

- We show that timestep-level conformal calibration is invalid for such policies and develop a patient-level blocked conformal risk control procedure that restores finite-sample control of the population alarm rate, with an explicit bound on the probability that it exceeds the budget.

- We propose an evaluation protocol that compares methods at matched alarm budgets, and we instantiate it on open multi-hospital cohorts under cross-hospital transfer, prevalence shift, and observation sparsification.

The remainder of the paper reviews related work (Section 2), develops the policy formulation and the calibration procedure (Section 3), and specifies the datasets, baselines, metrics, and experimental protocol (Section 4).

2. Related Work

2.1. Backbones for Irregularly Sampled Clinical Time Series

The modelling substrate for early warning is a sequence of observations whose sampling times carry as much information as their values. The most recent work in this line treats irregularity as a first-class object rather than an inconvenience: continuous-time formulations parameterise the latent trajectory between observations and evaluate it at arbitrary query times [8], reliability-aware aggregation weights observations by their estimated trustworthiness before temporal pooling [7], and graph constructions connect observations across variables and time so that forecasting can propagate information along non-uniform edges [9]. A complementary strand models the observation process itself, on the argument that the decision to order a laboratory test is a clinical action and therefore informative; this idea has been used both to synthesise realistic irregular trajectories [16] and to build masked-attention predictors for deterioration on general wards [10]. Domain-specific instantiations extend the same machinery to data-scarce settings such as venous thromboembolism detection after cancer treatment [17], and surveys of the area converge on the observation that architectural progress has outpaced agreement on evaluation [18]. We adopt these backbones without modification and treat the score function they produce as an input to our policy layer, which is where our contribution lies.

2.2. Early Warning and Deterioration Prediction

Clinical early warning has moved from static severity scores toward dynamic, trajectory-aware prediction. Recent multicentre work predicts the shape of sepsis deterioration rather than a single risk estimate, arguing that heterogeneous trajectories require patient-specific forecasting [1], while efficiency-oriented designs pursue compact models that remain explainable

under the compute constraints of a ward workstation [2]. Others inject clinical reasoning into the prediction pathway: dynamic temporal graphs embed calculator-derived features so that the model inherits the semantics of established instruments [12], and language-model-guided simulation is used to generate interpretable physiological dynamics for early warning [19]. Transfer-oriented formulations address the practical reality that a model trained in one institution must be reused in another with different coding practice and case mix [11], and earlier work established that deep models over longitudinal inpatient records can outperform tabular baselines for outcome prediction [3, 20, 21]. Across this literature the model is the unit of contribution; the alerting rule that converts its output into an operational signal is treated as an afterthought, and comparison between methods is almost always conducted at a fixed threshold rather than at a fixed alarm volume.

2.3. Calibration, Selective Prediction, and Conformal Guarantees

A parallel literature asks not whether a model is accurate but whether its outputs can be trusted at a stated level. Systematic analysis of calibration and selective prediction for multimodal clinical classification shows that abstention behaviour, not raw discrimination, determines whether a system is safe to deploy [13]. For time-series classifiers, reliability estimation has been extended beyond post hoc remapping of output scores toward evidence that a confident prediction is actually supported by the input [14], and audits of adaptation procedures have exposed failure modes that only appear when the update rule is evaluated with realistic label delays [22]. Conformal prediction supplies the distribution-free coverage guarantee that these approaches seek, and has been applied to physiological regression where calibrated uncertainty is a safety requirement [15]. Adaptation at inference time has also been used to counter patient-specific shift in haemodynamic prediction [23]. What this body of work assumes, and what an operational alarm system violates, is independence: a decision rule with a refractory period produces a dependent sequence, so guarantees derived from exchangeable examples do not transfer to the alarm rate actually experienced by a clinical unit. Our procedure is designed for exactly this dependent regime.

2.4. Evaluation Practice

Finally, the reporting conventions of the field shape what gets optimised. Risk-of-bias instruments for prediction models have recently been extended to cover artificial-intelligence-specific failure modes, including the handling

of calibration, heterogeneity, and operational applicability [24], and methodological reviews of intensive-care adverse-event prediction note that alert burden and time-to-alert remain inconsistently reported [4–6]. We take these critiques as a specification: every experiment below reports an operational quantity alongside a statistical one, and all comparisons between alerting strategies are conducted at matched alarm budgets so that discrimination is never bought with unreported alarm volume.

3. Method

3.1. Problem Setting and Notation

Let patient i contribute an observation set $\mathcal{O}_i = \{(t_{ij}, m_{ij}, v_{ij})\}_{j=1}^{n_i}$ with observation times $t_{ij} \in [0, T_i]$, variable indices $m_{ij} \in \{1, \dots, d\}$, and values $v_{ij} \in \mathbb{R}$. The set is irregular in the sense that neither the number of observations nor their spacing is controlled by the analyst: a laboratory panel may be drawn four times in an hour and then not for eight hours. A score function f_θ maps the history available at time t, denoted $\mathcal{O}_{i,\le t}$, to a risk score $s_{it} = f_\theta(\mathcal{O}_{i,\le t}) \in (0, 1)$. The score function is treated as a black box in this section; any of the backbones in Section 2 may be substituted, and Section 4 fixes a small set for the experiments.

The event of interest occurs at time $t_i^\star$, and for patients who never experience it we set $t_i^\star = \infty$. A prediction is useful if it fires in the window $[t_i^\star - W,\, t_i^\star)$ for a lead-time window W, which we fix at twelve hours following common practice in deterioration prediction. Because the score is defined at observation times only, all decisions below are indexed by the observation grid of each patient rather than by a global clock.

3.2. Stateful Alarm Policies

A threshold rule declares an alarm whenever $s_{it} \ge \lambda$. Such a rule has no memory, and in deployment it fires repeatedly on the same deteriorating patient, which is the mechanism by which alert volume becomes unmanageable. We therefore define an alarm policy $\pi_\lambda = (\lambda, \tau, \kappa)$ with three components: a threshold λ, a refractory window τ expressed in observation steps, and an escalation tier κ that specifies how many suppressed exceedances must accumulate before the refractory period is overridden. The policy carries an

internal state z_{it} and emits

$$a_{it} = \mathbb{1}[s_{it} \geq \lambda \ \wedge \ (z_{it} = 0 \ \vee \ c_{it} \geq \kappa)], \qquad z_{i,t+1} = \begin{cases} \tau, & a_{it} = 1, \\ \max(0,\, z_{it} - 1), & a_{it} = 0, \end{cases} \tag{1}$$

where c_{it} counts consecutive exceedances of λ that were suppressed. The refractory window is the control that directly trades alarm volume against repeated detection, and the escalation tier prevents a long refractory period from masking a genuinely escalating patient. Setting $\tau = 0$ and $\kappa = \infty$ recovers the memoryless threshold rule, so Equation (1) is a strict generalisation rather than a replacement.

Two properties of Equation (1) matter for what follows. First, the policy is a deterministic function of the score trajectory given (λ, τ, κ), so the only quantity that must be calibrated against deployment data is the scalar λ; the structural parameters are selected by cross-validation on the source cohort. Second, the emitted sequence $\{a_{it}\}$ is dependent in time by construction, and this dependence is not a nuisance to be averaged away but the mechanism through which the budget is enforced.

3.3. Why Timestep-Level Calibration Fails

It is tempting to calibrate λ by treating each observation time as an example and applying split conformal prediction to the indicator a_{it}. This is invalid for two distinct reasons, and separating them clarifies what the proposed procedure must repair.

The first is granularity. The operational quantity is a rate per patient-day,

$$R(\lambda) = \mathbb{E}\left[\frac{1}{D_i} \sum_{t} a_{it}(\lambda)\right], \tag{2}$$

where D_i is the length of patient i's record in days. The expectation is over patients, while the inner sum is over dependent observations. Controlling the marginal alarm probability at a randomly chosen timestep does not control Equation (2): the two differ by the inverse of the mean record length and by the within-patient correlation of a_{it}.

The second is state dependence. Because a_{it} depends on $a_{i,t-1}$ through z_{it}, the timestep-level losses are neither independent nor exchangeable once the policy is active. Calibrating on a pooled timestep sample therefore estimates the alarm rate of a policy evaluated on a different dependence structure from

the one that will run in the ward. The error is not conservative in either direction; it depends on how strongly alarms cluster, which is precisely the regime a well-designed refractory window creates.

3.4. Blocked Conformal Risk Control

The repair is to move the unit of calibration from the timestep to the patient. Define the patient-level loss

$$L_i(\lambda) = \frac{1}{D_i} \sum_t a_{it}(\lambda) \in [0,\, L_{\max}], \tag{3}$$

which is exactly the quantity inside the expectation in Equation (2). The bound is $L_{\max} = \sup_{H \geq H_{\min}} \frac{24}{H} \left(\left\lfloor \frac{H/\Delta - 1}{\tau + 1} \right\rfloor + 1 \right)$, where H is the length of stay, Δ the sampling interval and $H_{\min}$ the shortest stay admitted to the cohort. The supremum is not attained at one day: with hourly sampling and $\tau = 4$ a 26-hour stay permits six alarms, or $144/26 \approx 5.54$ per day, while a 24-hour stay permits only five. Evaluating the bound at a nominal day therefore understates it, and since short stays are exactly the ones that generate high per-day rates, $H_{\min}$ is a modelling decision that must be declared rather than a detail. Patients are exchangeable draws from the deployment population even though observations within a patient are not, so averaging Equation (3) over patients restores the exchangeability that conformal calibration requires. Let Λ be a finite grid of candidate thresholds and let $\widehat{R}_n(\lambda) = \frac{1}{n} \sum_{i=1}^{n} L_i(\lambda)$ be the calibration average over n held-out patients.

Proposition 1 (Budget validity)**.** *Fix $\alpha \in (0, 1)$ and a budget $B > 0$. Let*

$$\widehat{\lambda} = \inf \left\{ \lambda \in \Lambda : \; \widehat{R}_n(\lambda) + L_{\max} \sqrt{\frac{\log(|\Lambda|/\alpha)}{2n}} \;\leq\; B \right\}, \tag{4}$$

and set $\widehat{\lambda} = \max \Lambda$ if the set is empty. If calibration and deployment patients are exchangeable draws from the same population and L_i is bounded as in Equation (3), then

$$\Pr\left[R(\widehat{\lambda}) > B \right] \;\leq\; \alpha, \tag{5}$$

where $R(\lambda) = \mathbb{E}[L_i(\lambda)]$ is the population alarm rate defined in Equation (2).

Proof. For a fixed $\lambda \in \Lambda$, Hoeffding's inequality applied to the n bounded independent losses gives $\Pr[\widehat{R}_n(\lambda) - R(\lambda) \geq \epsilon] \leq \exp(-2n\epsilon^2/L_{\max}^2)$. A union bound over the $|\Lambda|$ candidate thresholds with $\epsilon = L_{\max}\sqrt{\log(|\Lambda|/\alpha)/(2n)}$ yields

$$\Pr\Big[\exists \lambda \in \Lambda : \ \widehat{R}_n(\lambda) - R(\lambda) \geq \epsilon\Big] \leq \alpha. \tag{6}$$

On the complement event, $\widehat{R}_n(\lambda) \geq R(\lambda) - \epsilon$ simultaneously for every λ. If $R(\widehat{\lambda}) > B$ then $\widehat{R}_n(\widehat{\lambda}) > B - \epsilon$, which contradicts the selection rule (4) that requires $\widehat{R}_n(\widehat{\lambda}) + \epsilon \leq B$. Hence $\{R(\widehat{\lambda}) > B\}$ is contained in the complement of the concentration event, whose probability is at most α. □

Three remarks follow. The bound is on the population alarm rate, not on the per-patient tail: a ward experiences both, and a guarantee about the mean does not imply that most patients stay within budget when the loss distribution is skewed. Section 4 therefore reports the fraction of patients above budget as a separate diagnostic and never presents it as a consequence of Proposition 1. Second, the slack scales as $L_{\max}\sqrt{\log|\Lambda|/n}$, so a finer grid buys operating-point resolution at a logarithmic cost, and the constant can be sharpened by replacing Hoeffding with an empirical-Bernstein bound, $\sigma(\lambda)\sqrt{2\log(2|\Lambda|/\alpha)/n} + 7L_{\max}\log(2|\Lambda|/\alpha)/(3(n-1))$, which is the variant used in all experiments and which reduces to a materially tighter threshold because per-patient alarm counts have low variance. Third, exchangeability across hospitals is an assumption that cross-site transfer violates; the cross-site experiments therefore report the empirical alarm rate rather than asserting that Equation (5) holds unconditionally.

Two implementation details follow from the proposition. Because $\widehat{R}_n$ need not be monotone on a finite sample, the selector is applied to an isotonic regression of $\widehat{R}_n$ on Λ; the certificate then refers to the smoothed risk, which is a pool-adjacent weighted average of neighbouring empirical risks rather than a single grid value, so the code exposes this as a switch and reports which variant was used. Second, the bound is vacuous if $L_{\max}\sqrt{\log(|\Lambda|/\alpha)/(2n)}$ approaches B, which happens for small calibration sets; the experiments therefore report the ratio of the realised alarm rate to the budget alongside the violation frequency, since a procedure that certifies the budget by emitting almost no alarms is valid but useless. The implementation also clips any loss above $L_{\max}$ and records how often that happened: for an admissible cohort the count is zero, and a non-zero count means $H_{\min}$ was declared too

Algorithm 1 Blocked conformal calibration of an alarm policy

Require: calibration patients $\{(\mathcal{O}_i, t_i^\star, D_i)\}_{i=1}^n$; grid Λ; budget B; level α; structural parameters (τ, κ)

for $\lambda \in \Lambda$ **do**
  run π_λ on each calibration patient, obtaining $\{a_{it}\}$
  $L_i(\lambda) \leftarrow D_i^{-1} \sum_t a_{it}$
  $\widehat{R}_n(\lambda) \leftarrow n^{-1} \sum_i L_i(\lambda)$
end for
$\widetilde{R}_n \leftarrow$ isotonic regression of $\widehat{R}_n$ on Λ (optional)
$\epsilon(\lambda) \leftarrow$ empirical-Bernstein slack of Proposition 1
$\widehat{\lambda} \leftarrow \inf\{\lambda : \widetilde{R}_n(\lambda) + \epsilon(\lambda) \le B\}$
return $\widehat{\lambda}$, and the empirical violation rate on a held-out split

generously, at which point the certificate no longer refers to the quantity that was measured.

3.5. Operating Point Selection and Lead Time

Given a calibrated policy, the remaining design question is which feasible threshold to use. Discriminative performance is not the right criterion, because all thresholds satisfying Equation (4) already meet the budget; among them the relevant quantity is how early the first alarm arrives relative to onset. For patient i with onset $t_i^\star$, define the lead time

$$\ell_i(\lambda) = \left(t_i^\star - \min\{t : \; a_{it}(\lambda) = 1, \; t < t_i^\star\}\right)^+, \tag{7}$$

and select λ within the feasible set to maximise the median of ℓ_i subject to a sensitivity floor, which is the constrained analogue of the usual clinical requirement that a screening rule not miss the majority of events. This is a two-stage procedure: Equation (4) certifies feasibility, and Equation (7) ranks the feasible operating points. Algorithm 1 states the complete procedure.

4. Experimental Setup

4.1. Datasets

All experiments use open, de-identified cohorts so that every reported number is reproducible without data-use agreements. Table 1 summarises

Table 1: Open cohorts used in this study. Alarm budgets are expressed per patient-day. The 2019 cohort provides three hospitals, which we index by the challenge's set identifiers.

Cohort	Patients	Variables	Task	Access
Challenge 2019, set A	20,336	40	Sepsis onset	Open
Challenge 2019, set B	20,000	40	Sepsis onset	Open
Challenge 2012, set A–C	12,000	42	Mortality	Open
MIMIC-IV demo	100	100+	Pipeline check	Open

the three sources. The primary cohort is the PhysioNet/Computing in Cardiology Challenge 2019 corpus, which contains hourly records from three independent hospitals: set A comprises 20,336 patients from one institution and set B comprises 20,000 patients from two further institutions. Because the three sites differ in coding practice, laboratory ordering policy, and case mix, the A$\rightarrow$B and A$\rightarrow$C directions constitute genuine cross-institution transfer rather than a random resplit of a pooled sample. Sepsis onset is defined by the Sepsis-3 criteria with the onset window spanning six hours before to three hours after the clinical diagnosis, as released with the challenge; because this labelling rule is itself a source of ambiguity, Section 4 specifies a sensitivity analysis over the window width.

The secondary cohort is the PhysioNet/Computing in Cardiology Challenge 2012 corpus, which provides a different task, in-hospital mortality, over a different variable set and a different era of care. It is used to test whether the calibration procedure depends on the specific label definition of the primary cohort. The MIMIC-IV demonstration subset is used only to verify that the preprocessing pipeline executes on an independently formatted source; with one hundred patients it is far too small to support any comparative claim and no result in this paper is derived from it.

4.2. Partitioning and Transfer Protocol

Every split is made at the patient level; no patient contributes observations to more than one partition. Within the source hospital we use a four-way partition of 60% training, 15% validation, 10% calibration, and 15% test. The validation partition selects early-stopping epochs and the structural policy parameters (τ, κ); the calibration partition is used only by Equation (4) and is never seen during training or model selection; the test partition is touched once per configuration.

Cross-site experiments follow a deployment-realistic protocol in two variants. In the *local-calibration* variant, the score function and structural parameters are fitted on the source hospital, and a small number $m \in \{50, 100, 200, 500\}$ of target-hospital patients is used for blocked conformal calibration before evaluation on the remaining target patients. This reflects the situation in which a site can supply a modest retrospective sample but not a labelled training cohort. In the *no-target-data* variant, calibration is performed entirely on source patients and transferred unchanged, which isolates how much of the guarantee survives when the exchangeability assumption is violated outright. The gap between the two variants is itself a reported result rather than a nuisance to be hidden.

4.3. Score Functions and Baselines

We separate the score function, which produces s_{it}, from the alarm policy, which consumes it. Holding the policy fixed while varying the score function isolates the contribution of the backbone; holding the backbone fixed while varying the policy isolates ours. The score functions are drawn from four families. Clinical instruments comprise the quick Sequential Organ Failure Assessment, the National Early Warning Score, the Modified Early Warning Score, the Systemic Inflammatory Response Syndrome criteria, and the full Sequential Organ Failure Assessment, computed from the released variables according to their published definitions. Classical learners comprise logistic regression and gradient-boosted trees over summary statistics of each patient's trajectory. Recurrent and convolutional networks comprise an long short-term memory network, a temporal convolutional network, and a residual network, implemented with the `tsai` time-series library. Irregularity-aware models comprise GRU-D and SAITS from the `PyPOTS` partially-observed time-series toolbox, together with reimplementations of reliability-aware temporal aggregation [7], masked-attention prediction with informative missingness [10], and graph-based forecasting over irregular observations [9].

Alarm policies comprise four baselines and the proposed policy. The memoryless threshold rule is the de facto standard and is tuned to the budget post hoc. The top-k rule alerts on the k highest-scoring timesteps per patient, with k chosen to meet the budget. The refractory-only rule applies Equation (1) with $\tau > 0$ but selects λ by grid search on the test alarm rate, which is the optimistic bound that a practitioner could not obtain prospectively. The timestep-conformal rule applies split conformal at the observation

level, which we include specifically to quantify the invalidity argued in Section 3. The proposed rule is Algorithm 1.

4.4. Metrics

Discrimination is reported as area under the receiver operating characteristic curve and area under the precision-recall curve, the latter because prevalence in the evaluation windows is low and the former is optimistic under imbalance. Operational quantities are alarms per patient-day, the false-to-true alarm ratio, the median and interquartile range of detection lead time from Equation (7), and the fraction of events detected at all within the twelve-hour window. Calibration is reported as expected calibration error with fifteen equal-width bins, the calibration slope, and reliability diagrams. Clinical utility is summarised by net benefit across threshold probabilities from 0.01 to 0.20, and by the official challenge utility score for the 2019 cohort. Budget validity is assessed on the quantity Proposition 1 actually bounds: the fraction of one hundred random calibration–test splits in which the held-out cohort's mean alarm rate exceeds B, reported against the nominal α with a Wilson interval. The per-patient violation rate, meaning the proportion of individual patients whose own per-day count exceeds B, is reported alongside it as a diagnostic and is never treated as a consequence of the guarantee. The two can move in opposite directions, and in the synthetic validation they do: sharper thresholds drive the mean rate down while pushing the per-patient tail up.

4.5. Statistical Analysis

Confidence intervals for all point estimates are obtained by bootstrapping patients with one thousand resamples. Differences in area under the curve between a pair of score functions are tested with the DeLong procedure for correlated curves; differences in operational metrics, which are not covered by DeLong, are tested with a paired Wilcoxon signed-rank test across the five training seeds, with the seed as the unit of pairing. Effect sizes are reported alongside exact p-values rather than as thresholded statements. Because one hundred calibration splits are available for the validity experiments, violation rates are accompanied by Wilson intervals, and a configuration is described as valid only when the interval contains the nominal level rather than when the point estimate falls below it. Subgroup analyses stratify by age band, sex, and comorbidity count, and are reported in full rather than filtered to significant contrasts.

4.6. Implementation and Compute

The full experimental grid comprises eight score functions $\times$ five seeds for the main comparisons, twelve baselines $\times$ three seeds for the policy comparison, and thirty ablation configurations. On a single 24 GB accelerator the entire programme requires approximately 45 accelerator-hours, dominated by the recurrent and irregularity-aware backbones; the conformal calibration stage is inference-only and contributes under three hours. Threshold grids use $|\Lambda| = 200$ equally spaced quantiles of the pooled calibration scores. Because $L_{\max}$ enters the slack directly, it is computed from the declared minimum stay $H_{\min}$ rather than assumed to equal one day: records shorter than twelve hours are excluded, which for hourly sampling and $\tau = 4$ gives $L_{\max} = 6$ instead of the 5.54 that a 24-hour supremum would wrongly suggest. Code, preprocessing scripts, and the exact patient-level partitions will be released on acceptance.

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